Hong Nhat - **Tran**

AI Engineer/Researcher · Bachelor's Degree

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"We can only see a short distance ahead, but we can see plenty there that needs to be done."

Summary

AI Research Engineer specializing in speech processing with hands-on experience in model quantization and collaboration workflows. Proven expertise in developing Vietnamese TTS systems, zero-shot voice cloning, and large-scale speech data pipelines. Seeking to leverage technology background and experience to contribute to AI initiatives in Vietnam.

Education

Hanoi University of Science and Technology
Hanoi, Vietnam

Master of Science in Computer Science — *2026-Present*

- GPA: 4.0

Hanoi University of Science and Technology
Hanoi, Vietnam

Bachelor of Computer Science — *2021-2026*

- CPA: 3.92
- Computer Science-Talent Program

Hanoi-Amsterdam High School for the Gifted
Hanoi, Vietnam

Physics-Specialized class — *2018 - 2021*

- English Proficiency: IELTS 8.0

Experience

Viettel AI & Data Services Center
Hanoi, Vietnam

AI Engineer (Previously Intern) — *Mar. 2024 - Present*

- Progressed from the Viettel Digital Talent 2024 internship program to a full-time role, receiving in-depth training in Data Science and Artificial Intelligence.
- Researched and developed state-of-the-art Vietnamese Text-to-Speech (TTS) systems, including single-speaker, multi-speaker, and Zero-shot Voice Cloning and speech editing.
- Built large-scale data collection pipelines for both ASR and TTS to continuously update and improve models with real-world data.
- Optimized and scaled deep learning models on high-performance infrastructure (NVIDIA DGX A100 and DGX B200), balancing performance with mobile AI deployment constraints.
- Collaborated with Qualcomm engineers on AI-driven speech edge-device product development, focusing on quantization workflows for Vietnamese TTS models.
- Developed a Speech/Telecommunication Channel Quality Assessment system for Vietnamese during the initial internship phase.

VinUni-Illinois Smart Health Center
Hanoi, Vietnam

Research Assistant — *Jan. 2022 - Dec. 2023*

- Work on projects about Classification of Pill datasets and Electrocardiography (ECG) data using Deep Learning Methods
- Support the writing and publication process of research papers in the team.

BKAI Lab, HUST
Hanoi, Vietnam

Project Team Member — *2021-2026*

- Work on projects about Classification of Pill datasets and Electrocardiography (ECG) data.

Projects

SpeechBakery
Viettel AI

Lead engineer/researcher/developer — *Aug. 2025 - Mar. 2026*

- Developed and implemented SpeechPie: a 7-step data pipeline for Vietnamese large-scale speech data processing from the Internet.
- Successfully crawled 10,000 hours of high-quality Vietnamese speech data for training large-scale Speech Synthesis models.
- Developed and implemented SpeechCake: a custom training and inference pipeline for Zero-shot Text-to-Speech, extended to voice design and paralinguistic generation for Vietnamese across 14 tags.
- Designed tests and metrics for benchmarking Vietnamese Voice cloning systems.
- The resulting model reached SOTA Voice Cloning and Speech Editing performance compared to all Vietnamese Text-to-Speech systems.

Personal AI

Viettel AI

AI Engineer/Researcher — *Mar. 2026 - Now*

- Developing low-latency on-device Text-to-Speech and Zero-shot TTS applications for Viettel's Personal AI phone, built on Qualcomm's reference design and SDK.
- Adapting English TTS, Vision-Language (VLM) and embedding models to Vietnamese tasks and applications.
- Optimizing models for on-device inference under Qualcomm's SDK constraints.
- Researching phone agents and UI agents to enable on-device agentic automation.

Vietnamese Speech-to-Speech Agent

Viettel AI

Researcher/developer — *Jun. 2026 - Now*

- Architecting the framework for an end-to-end cascaded system integrating a Speech Language Model with a Controllable TTS pipeline.
- Reverse-engineered and finetuned Qwen3-TTS for controllable speech synthesis (voice design, custom voice generation, voice cloning.)
- Prototyping and curating Vietnamese initial training datasets to establish proof-of-concept for controllable speech synthesis features.

Vietnamese Crosslingual Text-to-Speech

Viettel AI

Completed (Artificial Intelligence, Natural Language Processing) — *Aug. 2024 - Nov. 2024*

- Used pretrained Zero-shot Text-to-Speech models to generate high-quality Cross-lingual Vietnamese-English data.
- Trained high-quality lightweight Vietnamese TTS models capable of generating English borrowings
- Trained Neural Vocoders across 10+ speakers, designed TTS datasets and benchmarks for training and evaluation.

Telecommunication Channel Quality Assessment

Viettel AI

Research Engineer — *Mar. 2024 - Jun. 2024*

- Developed a deep learning model for automatic telecommunication channel quality assessment
- Developed benchmarks, metrics and documents for the project

VAIPE

VinUniversity

Research Assistant — *Mar. 2023 - Dec. 2023*

- Researched and implemented Deep Learning Algorithms for detecting cardiac irregularities through electrocardiogram signals.
- Supported the process of pill/precription datasets labeling.
- https://vaipe.health/

Publications

Paper: H. N. Tran, T. L. Nguyen, L. T. D. Nguyen, N. M. Le, V. H. Do, "TTS^2: Test-Time Scaling for Flow-Based Text-to-Speech via Multi-stage Reward Modeling on Posterior Mean Projections," *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

Paper: D. S. Dinh, T. M. A. Nguyen, **H. N. Tran**, T. T. H. Ngo, "LitCodec: ASR-Guided Streaming Speech Coding with Unified Quantization" Proceedings of the 27th Annual Conference of the International Speech Communication Association (INTERSPEECH 2026)

Thesis: H. N. Tran, "SpeechBakery: A unified pipeline for Vietnamese speech collection and Zero-shot Text-to-Speech synthesis," Graduation Thesis, Hanoi University of Science and Technology, 2026.

Paper: H. N. Tran, T. L. Nguyen, T. H. Tran, N. M. Le, and V. H. Do, "Efficient Caching for Conditional Flow Matching in Vietnamese Zero-Shot TTS," Symposium on Information and Communication Technology (SOICT), 2025.

Paper: H. N. Tran, B. T. Ta, and V. H. Do, "Vietnamese Speech Database for No-Reference Telecommunication Quality Assessment," 28th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA), IEEE, 2025.

Paper: H. N. Tran, T. L. Nguyen, and T. H. Tran, "ViettelRoar: Voice Conversion approach for VLSP 2025," INLG Workshop, 2025 (Second Prize).

Paper: M. T. Cao, **H. N. Tran**, P. L. Nguyen, T. H. Nguyen, and H. H. Pham, "IncepSE: Leveraging InceptionTime's performance with Squeeze and Excitation mechanism in ECG analysis," Symposium on Information and Communication Technology (SOICT), 2023.

Paper: M. T. Cao, **H. N. Tran**, P. L. Nguyen, T. H. Nguyen, and H. H. Pham, "Multimodal contrastive learning for diagnosing cardiovascular diseases from electrocardiography (ECG) signals and patient metadata," Midwest Machine Learning Symposium (MMLS), 2023.

Honors & Awards

2025	**Outstanding Employee,** Voice core team	*Viettel AI*
2025	**Second prize,** Vietnamese Voice Conversion Challenge	*VLSP-2025*
2021	**Merit Scholarship,** Four semesters	*HUST*
2019	**Third prize,** National Physics Olympiad	
2019	**First prize,** Hanoi Physics Olympiad	